Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | linkedin.com/in/varunprasaath-rs |

PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in polymer materials, organic chemistry, coatings, characterization, DOE/SPC, and process scale-up. Currently leading bio-based polymer process development and translating FT-IR, TGA/DSC, NMR/GPC, UV-Vis, and failure-analysis data into reproducible specifications.

CORE COMPETENCIES

Polymer Development | Organic & Polymer Chemistry | Coatings and Non-Metallic Materials | DOE / SPC | Root-Cause Failure Analysis | FT-IR / TGA / DSC / GPC / NMR | Process Scale-Up | Technical Documentation

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, converting material-quality data into reproducible process windows, scale-up specifications, and SOP-ready controls.
- Define polymer characterization readouts and technical documentation that support formulation decisions, manufacturability, and quality-relevant specifications.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Assessed reliability risks in functional organic materials by identifying degradation, delamination, and redox-instability failure modes across 15+ variants using spectroscopy, electrochemistry, thermal analysis, NMR/GPC, and MS.
- Developed structure-property insights for organic material systems by linking synthesis, purification, characterization, and performance data.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded thin-film coating process development, linking CVD/surface preparation to morphology, interface quality, optical performance, and process reproducibility.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Developed and optimized CVD polymer thin-film deposition with DOE strategies and in-situ QCM-D monitoring to define growth kinetics and reproducible process controls.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Bio-Based Polymer Scale-Up Process Current

Defined DOE-guided process windows, material-quality specifications, and SOP-ready controls for polymer systems moving toward scale-up.

DOE, SPC, FT-IR, thermal analysis, process documentation

AFOSR Defense Thin-Film Coatings DoD Funded

Established process-property relationships for CVD/surface-preparation workflows and reproducible coating conditions.

CVD, QCM-D, surface prep, polymer thin films

Electrochromic Material Reliability Study Chemistry of Materials 2025

Mapped degradation and failure modes across functional organic materials using orthogonal characterization and stress-test logic.

FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC, MS

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Polymer & Non-Metallic Materials: Bio-based polymers, CVD polymer thin films, coatings, organic materials, surface functionalization, structure-property-processing relationships
Characterization: FT-IR, UV-Vis-NIR, ellipsometry, GPC/SEC, LC-MS, GC-MS, NMR, QCM-D, TGA, DSC, cyclic voltammetry
Development & Scale-Up: DOE, SPC, process windows, formulation/process variables, reproducibility studies, SOPs, scale-up specifications, technical reports
Failure Analysis: Failure mode identification, root cause analysis, degradation analysis, anomalous data investigation, corrective-action recommendations
Data & Communication: Excel, Python/basic, JMP/basic, technical presentations, publication-quality writing, cross-functional technical updates

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)